

Emil Stroecker

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EDUCATION

Maastricht University

Research MSc in Cognitive Neuroscience — (current GPA: 8.8/10.0 → US: 4.0 /4.0)

Maastricht, Netherlands
09/2024 – 08/2026

- Selective 2-year program with research focus, featuring 10-month MSc thesis research
- *Selected coursework:* Statistics I-III, Applied Statistics I-II, Basic Mathematical Methods, Signal Analysis, MATLAB Advanced Programming, Python Programming, Quantificational Logic and Deductions, Advanced fMRI, EEG & MEG, Brain Connectivity & Connectomics, Sensorimotor Processing, Functional Neuroanatomy

Kyoto University

Kyoto University International Exchange Program — (US GPA 4.0/4.0)

Kyoto, Japan
09/2023 – 02/2024

- Received [JASSO scholarship](#)

Maastricht University

BSc in Psychology — Summa Cum Laude, Top 1% (GPA: 9.1/10.0 → US: 4.0/4.0)

Maastricht, Netherlands
09/2021 – 08/2024

- Empirical programme with a focus on research methods, statistics, cognitive psychology, and neuroscience

RESEARCH EXPERIENCE

Columbia University, Zuckerman Institute

MSc Thesis Research Intern at Nikolaus Kriegeskorte's [Visual Inference Lab](#)

New York City, USA
11/2025 – 08/2026

- Acting as project lead: researching neural network models of ventral stream and their bias towards geometrical interpretations of depth in the Ames Window Illusion
- Comparing contemporary neural networks (Metric3D, DSINE, MoGe) for depth estimation
- Running psychophysics experiment (PsychoPy/PsychoJS) for benchmarking
- Creating online visual perception games for large-scale data collection
- Experimental documentation used as foundation for AI psychophysics benchmark tool ("illugen")
- First author for 2-page abstract, accepted to Conference on Cognitive Computational Neuroscience 2026

Maastricht University

Research Assistant at [Developmental Cognitive Neuroscience Group](#)

Maastricht, Netherlands
09/2025 – 11/2025

- Refactored and debugged a [psychophysics experiment](#) in PsychoPy for EEG compatibility and improved task paradigm

Independent Researcher with Rainer Goebel on fMRI decoders

09/2025 – 12/2025

- Decoded representations of viewed and imagined letters in V1 from fMRI voxel patterns
- Implemented GCN, CNN, and FCNN in PyTorch as a proof-of-concept for comparing decoding architectures

Sokendai, National Institute for Physiological Sciences

Internship in primate research at Ryo Sasaki's [Multisensory Integration Systems Lab](#)

Okazaki, Japan
07/2025

- Comprehensive training on major practical aspects of macaque experiments
- Observed monkey experiments full-time for two weeks, witnessing full procedures for behavioural training & neural recordings for saccade tasks & spatial navigation tasks

Maastricht University

BSc Honours Thesis (MaRBLc) under Vincent van de Ven

Maastricht, Netherlands
11/2023 – 07/2024

- Fully self-directed project solely responsible for entire experiment with only minimal supervision: experimental design, experiment creation (PsychoPy/PsychoJS), recruiting, administration to 40 participants, analysis, etc.
- Examined effect of structural association of tones on tone sequence recognition
- Consistent with literature, musicians demonstrated significantly better recognition of structured tone sequences
- Presented final poster at symposium to faculty and fellow honours students, project graded outstanding (10/10)

Interdisciplinary Research Project through [Honours+](#)

11/2022 – 05/2023

- Self-directed, interdisciplinary, empirical research, selecting top 5% of BSc students across disciplines
- Led a team of 5 honours students: conducted a qualitative content analysis on TikTok content, laying foundation for quantitative cross-sectional study with 100+ participants; findings presented to faculty

University Hospital Cologne

Research Intern in TrainParC-Advanced research project

Cologne, Germany
07/2023 – 09/2023

- Conducted and evaluated various neuropsychological & motor tests (MMST, BDI-II, MoCA, FOG-Q, etc.) with Parkinson's patients
- Developed migration from paper to digital data entry pipeline in Excel, significantly reducing workflow time
- Contributions acknowledged in a resulting Springer Nature Link publication ([Trials, 2025](#))

PRESENTATIONS

Conference Abstracts

Stroecker, E., Cheng, F., Kriegeskorte, N. (2026, August). Measuring Depth Inversion in the Ames Window Illusion [Extended abstract, accepted].
9th Annual Conference for Cognitive Computational Neuroscience, NYC, USA. [[Link](#)]

Posters

- Stroecker, E., van de Ven, V. (2024, July 05). *Measuring on-line predictive recall of structured versus non-structured tone sequences*. [Poster presentation]. *15th Marble Symposium*, Maastricht, The Netherlands. [\[Link\]](#)
- Stroecker, E., Kud, H., Husarcikova, V., van Kuilenburg, K., Eckstein, H. (2023, May 10). *The Impact of TikTok Consumption on the Perception of ADHD*. [Poster presentation]. *Honours+ Closing Symposium*, Maastricht, The Netherlands. [\[Link\]](#)

HONOURS & AWARDS

Honours Programme, PREMIUM — extracurricular business consultant for Atos	01/2025 – 05/2025
Summa Cum Laude Designation, BSc Psychology (= top 1% of the class)	08/2024
Honours Programme, MaRBL (selective for top 20%) — empirical thesis research project	11/2023 – 07/2024
Scholarship, Japanese Student Services Organization (JASSO) — 400,000JPY = 2,500USD	10/2023 – 02/2024
Honours Programme, Honours+ (selective for top 5%) — extracurricular research project on societal topics	11/2022 – 05/2023
Honours Programme, FPN Honours Programme (selective for top 5%) — extracurricular courses beyond Psychology	09/2022 – 09/2023

ACADEMIC SERVICE / OTHER

In-Mind Magazine (in-mind.org)	remote
Editorial Assistant for Editorial Management Team	05/2026 – ongoing
- Spearheaded development of editorial guidelines for incoming, peer-reviewed submissions - Screening submissions to ensure compliance with guidelines	
Columbia University	New York City, USA
Research Mentor for BUMP Biology	01/2026 – 12/2026
Advising undergraduate student on first steps into neuroscientific research, i.e. applying for research programmes, contacting PIs, shaping academic goals & development, etc.	
9th Annual Conference on Cognitive Computational Neuroscience (CCN2026)	New York City, USA
Workflow Lead for Technical Program Committee	10/2025 – 08/2026
- Managing and facilitating submission process for 600+ submissions on OpenReview - Documenting and deciding conference policies and procedures, shaping future iterations of the venue	
Maastricht University	Maastricht, Netherlands
Reviewer for Maastricht Student Journal of Psychology and Neuroscience (MSJPN)	02/2025 – ongoing
Teaching Assistant for BSc courses	11/2024 – 07/2025
- Instructed tutorial group meetings and conveyed course content to 60+ students - Courses: Functional Neuroanatomy, Statistics, Consciousness, Psychodiagnostics	

WEB ARTICLES

BrainMatters (brainmatters.nl)	remote
Writer — neuroscience-related articles intended for non-expert audiences	04/2025 – ongoing
<u>Stroecker</u>, E. (2026, June 22). <i>Beauty and the Brain</i>. BrainMatters. [Link] <u>Stroecker</u>, E. (2025, October 18). <i>fNIRS</i>. BrainMatters. [Link] <u>Stroecker</u>, E. (2025, September 30). <i>The revolutionary finding that changed the meaning of functional magnetic resonance imaging forever</i>. BrainMatters. [Link] <u>Stroecker</u>, E. (2025, May 29). <i>How Brain Reading is Becoming Reality Through Real-Time fMRI</i>. BrainMatters. [Link]	
Maastricht University , FPN Honours Programme	Maastricht, Netherlands
- Conducted historical book review (100+ years old), on book's content, author's biography, and historical context, culminating in a dedicated Wikipedia page: <i>The Myths and Legends of the North American Indians</i>. (2023, December 08). In <i>Wikipedia</i>. Retrieved February 24, 2026, from [Link]	
	12/2023

SKILLS

Languages

German (native) | English (professional) | Japanese (intermediate) | Dutch, French, Spanish (elementary)

Applied Skills

Languages:	Python, Bash, Matlab, R, JavaScript, SQL, HTML/CSS
ML & Data:	PyTorch, NumPy, Pandas, Scikit-learn, Matplotlib
Tools:	PsychoPy, Blender, LaTeX (Overleaf), GitHub, MS Office, Qualtrics, OpenReview
Neuroimaging:	fMRI paradigms (real-time fMRI, neurofeedback, ...), EEG, NIBS (TMS, tDCS,...), electrode recording procedures
Other:	Psychophysics, statistical & logical inference, writing

Trainings & Certifications

TOEFL iBT 118/120 ⇔ 6.0/6.0 (equivalent to CEFR C2) | Columbia University ZI fMRI Certification (L2) |

UM Honours Professional & Personal Workshops (10+)